Averaged response to stimuli and detection of visually responsive sites

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1 Load previously saved data

The cleaned stimuli (with repetition number) and the session-specific block orders can be read from `data/processed/20260521`. The MUAs saved with `qs2` are saved in `data/processed/20260521/mua`.

2 PSTHs between developmental stages and region recorded

2.1 Original Lee et al. strategy

Within the MATLAB files, Gerrick Lee writes:

> *In the manuscript, I combined data for analysis purposes. Figure 3A[1] contains PSTHs from 10 example sites, taken from "stimuli presented in the first position of a stimulus block. To combine data across recordings, the correlations between sites within a recording have to be shuffled - in doing this, the position wrt the stimulus block is lost. Hence, the PSTHs don't exactly match those in Fig 3A. This script takes a subset of the original data (the ten example sites), and plots the PSTHs from the manuscript.*

Though the file `psth_data.mat` contains the data needed to re-plot Figure 4A, I could not find the selection procedure for the sites. The `psth_data.mat` file also contains more than 4600 stimulus labels for each session, which is inconsistent with the other data. Furthermore, for sessions with multiple subjects, the session file only contains one vector of block order. I am unsure if all subjects had stimuli presented in identical order or if data is missing. I will tentatively assume the data is complete, as the block order is described as "pseudorandom" in the first paragraph under "Physiological experiments" (p. 14). If needed, I will reach out to Gerrick Lee directly for formatting questions, though I doubt it is really that important.

2.2 Attempted reconstruction

I attempted to reproduce the PSTHs by arbitrarily selecting ten sites in each subject and averaging across rows with `block_order` equal to `1`. Unlike Lee et al., I averaged separately by subject, when

[1] The published version of Lee et al. displays PSTHs in Figure 4A, not 3A.

applicable. The selected sites were probably not randomly sampled but instead only taken from the visually responsive sites, explaining the discrepancy between my reconstruction and Figure 4A.

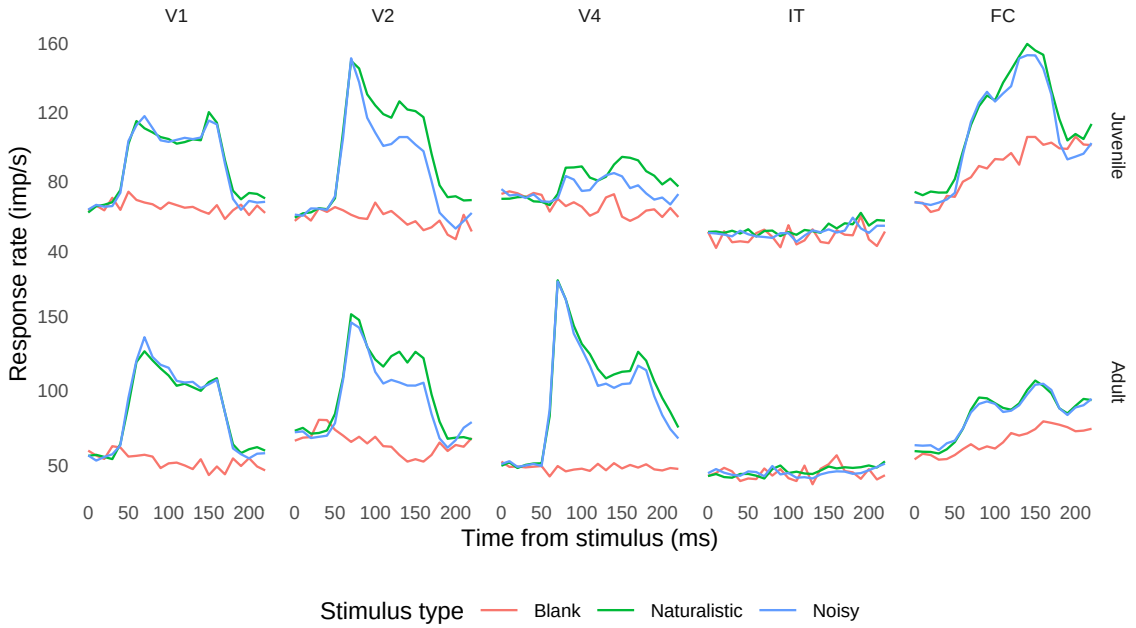


Figure 1: Averaged MUA for regions and the youngest (juvenile) and oldest (adult) recording ages, which may differ by region.

3 Visually responsive sites and onset time

Within the supplement of the main paper, Lee et al. writes:

> *To determine whether a site was visually responsive in a given recording session, we measured the average response separately for even and odd repetitions of each stimulus. We measured the Pearson correlation between these vectors, and used sites with a correlation of 0.4 or greater (the results reported here remained stable across a variety of different metrics and thresholds)…We defined the onset time for a given site as the first deviation above a d' of ± 0.4, as long as the following 2 windows were also suprathreshold (and of the same sign). The onset time for a given population was the median onset time across visually responsive sites. For most analyses, we then summed responses across the 150 ms window following stimulus onset.*

The d' referenced is Cohon's d, calculated as

$$d' = \frac{\mu_1 - \mu_0}{\sqrt{(\sigma_1^2 + \sigma_0^2)/2}},$$

where contrasting blanks and stimuli provides the d' for *sensitivity* and contrasting noisy and naturalistic stimuli provides the d' for *selectivity*.

The procedure for finding onset site is a bit circular.

Table 1: Number of responsive sites found for each region and age recorded, with arrays numbered arbitrarily.

Age (wks)	Array 1	Array 2	Array 3	Total sites
V1				
29	21	NA	NA	21
37	23	0	NA	23
56	34	NA	NA	34
V2				
29	19	58	NA	77
37	21	49	NA	70
56	16	45	NA	61
V4				
29	39	3	NA	42
37	43	15	NA	58
56	41	27	NA	68
409	77	NA	NA	77
IT				
20	13	NA	NA	13
31	2	4	2	8
34	4	2	NA	6
221	11	NA	NA	11
FC				
20	19	NA	NA	19
31	4	NA	NA	4
35	0	NA	NA	0

1. Find a temporary onset time, with d' calculated between blanks and stimuli. The search window is 0 to 200 ms after stimulus presentation. If no onset time is found, use 100 ms as a placeholder.
2. Determine the responsive sites with signals averaged over a 150 ms window starting at the temporary onset time.
3. Find the onset time based on the responsive sites.

For reference, see line 96 in `repository_textures2_analysis_figures_v0.m`, which calls `Textures2_onset_via_discrim.m`. The responsive sites found with the temporary onset time calculation can be seen in Table 1. The final onset time is then calculated from only the responsive sites and grouped by region and age, seen in Table 2.

My results may differ from Lee et al. because in Step 1, I calculate a temporary onset time separated by array/subject. In contrast, Lee et al. performs all calculations after concatenating recordings from the same region and age, which may contain up to three different arrays.

Table 2: Calculated onset times, when responsive sites are grouped by region and age. FC at 35 weeks has no responsive sites and therefore defaults to an onset of 100 ms in future calculations. The last column reports the proportion of sites where an onset was found.

Age (wks)	Onset time (ms)	Min. onset	Max. onset	Responsive sites	Prop. with onset
V1					
29	40	60	70	21	1.00
37	40	60	70	23	0.957
56	40	60	80	34	1.00
V2					
29	60	60	130	77	0.987
37	60	60	80	70	1.00
56	60	70	130	61	0.984
V4					
29	70	80	170	42	0.810
37	100	80	150	58	0.431
56	70	70	140	68	0.471
409	60	70	80	77	0.987
IT					
20	140	160	160	13	0.0769
31	110	120	150	8	0.375
34	90	110	170	6	0.667
221	90	110	110	11	0.0909
FC					
20	70	70	130	19	1.00
31	80	70	120	4	1.00
35	NA	NA	NA	0	0

4 Averaged responses

The responsive sites and onset times produce the time-averaged matrices of the responses. I grouped measurements by region and age, i.e., removing the distinction between subjects, and averaged responses in a 150 ms window.